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EXP.NO:4 Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

CODE:

```
import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.utils import to_categorical
iris = load_iris()
X = iris.data
y = iris.target
y = to_categorical(y)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = Sequential()
model.add(Dense(8, activation='relu', input_shape=(4,)))
model.add(Dense(3, activation='softmax'))
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
model.fit(X_train, y_train, epochs=20, batch_size=5, verbose=0)
loss, accuracy = model.evaluate(X_test, y_test, verbose=0)
print("Test Accuracy: {:.2f}%".format(accuracy * 100))
sample = np.array([[5.1, 3.5, 1.4, 0.2]])
sample = scaler.transform(sample)
```

```
prediction = model.predict(sample, verbose=0)
predicted_class = np.argmax(prediction)
print("Predicted Class:", iris.target_names[predicted_class])
```

OUTPUT:

```
Test Accuracy: 96.67%
Predicted Class: setosa
```